

Spectra and Pseudo-Spectra

We now focus on the spectral theory of bounded endomorphisms, i.e. operators

$$T \in \mathcal{B}(X, X).$$

A fundamental tool for the definition of the spectrum is the resolvent,

$$R_T(z) := (T - zI)^{-1}, \quad z \in \mathbb{C}.$$

The resolvent might not be well-defined on the whole complex plane. We will drop the notation $\|\cdot\|_X$ and write simply $\|\cdot\|$.

1 Spectrum and resolvent

Definition 1.1. Given an operator $T \in \mathcal{B}(X, X)$, we call the *resolvent set* of T the subset of $\mathbb{C}$ on which $R_T(z)$ is well-defined and belongs to $\mathcal{B}(X, X)$, i.e.

$$\rho_T := \{z \in \mathbb{C} : T - zI : X \rightarrow X \text{ is bijective}\}.$$

We call the *spectrum* of T the complement of the resolvent set,

$$\sigma_T := \mathbb{C} \setminus \rho_T.$$

Proposition 1.2. Let $T \in \mathcal{B}(X, X)$. Then ρ_T is open and therefore σ_T is closed.

Proof. Let $z \in \rho_T$ and pick $z' \in \mathbb{C}$ such that

$$|z - z'| < \|R_T(z)\|^{-1}.$$

Then

$$T - z'I = (T - zI) + (z - z')I.$$

The perturbation $S = (z - z')I$ satisfies

$$\|S\| = |z - z'| < \|R_T(z)\|^{-1}.$$

By the first stability estimate, $T - z'I$ is bijective and

$$R_T(z') \in \mathcal{B}(X, X).$$

Thus $z' \in \rho_T$, so ρ_T is open. Hence $\sigma_T = \mathbb{C} \setminus \rho_T$ is closed. □

Proposition 1.3. Let $T \in \mathcal{B}(X, X)$. Then the spectrum of T is non-empty, i.e.

$$\sigma_T \neq \emptyset.$$

Proof. Assume by contradiction that $\sigma_T = \emptyset$. Then $\rho_T = \mathbb{C}$, so $R_T(z)$ is analytic on all of $\mathbb{C}$.

For $|z| > \|T\|$, the Neumann series gives

$$(T - zI)^{-1} = -\frac{1}{z} \left(I - \frac{T}{z} \right)^{-1} = -\sum_{k=0}^{\infty} \frac{T^k}{z^{k+1}}.$$

Therefore

$$\|R_T(z)\| \rightarrow 0 \quad \text{as } |z| \rightarrow \infty.$$

Since R_T is analytic on $\mathbb{C}$ and bounded at infinity, it is bounded on $\mathbb{C}$. By Liouville's theorem, R_T is constant. Since $\|R_T(z)\| \rightarrow 0$ as $|z| \rightarrow \infty$, this constant must be zero, which is impossible because $R_T(z)$ is an inverse operator. Hence $\sigma_T \neq \emptyset$. $\square$

Theorem 1.4. *Let $T \in \mathcal{B}(X, X)$ and $z \in \rho_T$. Then*

$$\|R_T(z)\| \geq \text{dist}(z, \sigma_T)^{-1}. \quad (*)$$

Proof. By the previous proof, if

$$|z - t| < \|R_T(z)\|^{-1},$$

then $t \in \rho_T$. Equivalently, if $t \in \sigma_T$, then

$$|z - t| \geq \|R_T(z)\|^{-1}.$$

Taking the infimum over $t \in \sigma_T$ gives

$$\text{dist}(z, \sigma_T) \geq \|R_T(z)\|^{-1},$$

which is the desired inequality. $\square$

Proposition 1.5. *If $z \notin \sigma_T$, then $z \notin \sigma_{T+S}$ for every $S \in \mathcal{B}(X, X)$ such that*

$$\|S\| \leq \|R_T(z)\|^{-1}.$$

Proof. This is a direct consequence of the proof that ρ_T is open and of the first stability estimate. $\square$

The previous proposition asserts the upper semi-continuity of the spectrum: an infinitesimal perturbation of T can move the spectrum only infinitesimally.

2 Pseudo-spectra

We now give a definition of pseudo-spectra based on estimate (*), analogous to the one of Trefethen.

Definition 2.1. Let $T \in \mathcal{B}(X, X)$ and fix $\varepsilon > 0$. We denote by $\sigma_{\varepsilon, T}$ the ε -pseudo-spectrum of T , defined by

$$\sigma_{\varepsilon, T} := \left\{ z \in \rho_T : \|R_T(z)\| > \frac{1}{\varepsilon} \right\} \cup \sigma_T.$$

Equivalently, the complement

$$\sigma_{\varepsilon, T}^c = \left\{ z \in \rho_T : \|R_T(z)\| \leq \frac{1}{\varepsilon} \right\}$$

is called the *pseudo-resolvent*. Fixing a level of stability that we are willing to tolerate, the pseudo-resolvent is the set of $z \in \mathbb{C}$ for which we can control sensitivity to perturbations, namely for which

$$\|x\| = \|R_T(z)y\| \leq \frac{1}{\varepsilon} \|y\|.$$

This means that small data produce small solutions, and small variations in the data can cause only small variations of the solution. It does not mean that a small perturbation of the solution corresponds to a small perturbation of the data.

Example 2.2 (Helmholtz-type equation). Consider a Helmholtz-type equation

$$(T - zI)x = y, \quad z \in \rho_T.$$

Since $z \in \rho_T$, for each $y \in X$ there exists a unique solution

$$x = R_T(z)y.$$

Thus

$$\|x\| \leq \|R_T(z)\| \|y\|.$$

The previous spectral estimate gives only a lower bound on $\|R_T(z)\|$, so in general it does not control the norm of the solution. The situation is very different when the operator is normal. For non-normal operators, the lack of control on $\|R_T(z)\|$ suggests that the Helmholtz problem can be ill-conditioned: a small perturbation of the data can produce a completely different solution.

This motivates the physical significance of non-normal operators. In the words of Nick Trefethen:

If your answer is highly sensitive to perturbations, you have probably asked the wrong question.

In this sense, the complement of the pseudo-spectrum is an especially interesting set.

Proposition 2.3. *Let $T \in \mathcal{B}(X, X)$. If $\varepsilon_1 > \varepsilon_2 > 0$, then*

$$\sigma_{\varepsilon_2, T} \subseteq \sigma_{\varepsilon_1, T},$$

and

$$\bigcap_{\varepsilon > 0} \sigma_{\varepsilon, T} = \sigma_T.$$

Furthermore, each connected component of $\sigma_{\varepsilon, T}$ has non-empty intersection with σ_T . In particular, $\sigma_{\varepsilon, T}$ is closed.

Proof. The monotonicity follows directly from the definition. Indeed, if $\varepsilon_1 > \varepsilon_2$, then

$$\frac{1}{\varepsilon_1} < \frac{1}{\varepsilon_2},$$

so the condition $\|R_T(z)\| > 1/\varepsilon_2$ implies $\|R_T(z)\| > 1/\varepsilon_1$.

The identity

$$\bigcap_{\varepsilon > 0} \sigma_{\varepsilon, T} = \sigma_T$$

follows because, if $z \in \rho_T$, then $\|R_T(z)\| < \infty$, so for sufficiently small ε we have $\|R_T(z)\| \leq 1/\varepsilon$, and hence $z \notin \sigma_{\varepsilon, T}$.

To prove the statement about connected components, observe that R_T is analytic on ρ_T , hence $\|R_T(z)\|$ is a subharmonic function. If a connected component of $\sigma_{\varepsilon, T}$ did not intersect σ_T , then the maximum principle for subharmonic functions would force the maximum of $\|R_T(z)\|$ to be attained on the boundary, contradicting the definition of the component. Therefore every connected component of $\sigma_{\varepsilon, T}$ intersects σ_T . $\square$

3 Eigenvalues and spectral radius

Definition 3.1. We call $z \in \mathbb{C}$ an *eigenvalue* of $T \in \mathcal{B}(X, X)$ if

$$\mathcal{N}(T - zI) \neq \{0\}.$$

We denote the set of eigenvalues of T by Σ_T . If $z \in \Sigma_T$, we call $\mathcal{N}(T - zI)$ the eigenspace corresponding to z .

Proposition 3.2. *Let $T \in \mathcal{B}(X, X)$. Then*

$$\Sigma_T \subseteq \sigma_T.$$

Proof. If $\mathcal{N}(T - zI) \neq \{0\}$, then $T - zI$ is not injective. Therefore $T - zI$ is not bijective, so $z \in \sigma_T$. $\square$

It may happen that

$$\mathcal{N}(T - zI) = \{0\} \quad \text{and} \quad \mathcal{R}(T - zI) \neq X.$$

In this case $z \in \sigma_T$ but $z \notin \Sigma_T$.

Example 3.3 (Right-shift operator). Let $X = \ell^2$ and consider the right-shift operator

$$T : X \rightarrow X, \quad T(u_1, u_2, u_3, \dots) = (0, u_1, u_2, u_3, \dots).$$

Then

$$Tu = \lambda u$$

means

$$(0, u_1, u_2, \dots) = (\lambda u_1, \lambda u_2, \lambda u_3, \dots).$$

Hence

$$\lambda u_1 = 0, \quad \lambda u_2 = u_1, \quad \lambda u_3 = u_2, \dots$$

and the only possible eigenvector is the zero vector. Therefore

$$\Sigma_T = \emptyset.$$

However, $\sigma_T \neq \emptyset$ for every bounded operator, so in this example

$$\Sigma_T \subsetneq \sigma_T.$$

Definition 3.4. Given an operator $T \in \mathcal{B}(X, X)$, we call the *spectral radius* of T the number

$$r_T := \sup_{z \in \sigma_T} |z|.$$

Theorem 3.5 (Gelfand formula). *Let $T \in \mathcal{B}(X, X)$. Then*

$$r_T = \lim_{n \rightarrow \infty} \|T^n\|^{1/n}.$$

Corollary 3.6. *Let $T \in \mathcal{B}(H, H)$ be self-adjoint. Then*

$$r_T = \|T\|.$$

Proof. If T is self-adjoint, then

$$\|T^2\| = \|T\|^2.$$

Thus

$$\|T^{2^n}\|^{1/2^n} = \|T\|.$$

Using the Gelfand formula gives $r_T = \|T\|$. □

Corollary 3.7. *Let $T \in \mathcal{B}(H, H)$ be self-adjoint. If*

$$\sigma_T = \{0\},$$

then $T = 0$.

Proof. If $\sigma_T = \{0\}$, then $r_T = 0$. Since T is self-adjoint,

$$\|T\| = r_T = 0,$$

so $T = 0$. □

Definition 3.8. Let $T \in \mathcal{K}(X, X)$ and let $z \in \Sigma_T \setminus \{0\}$. We define the *ascent* of z as the smallest $\alpha \in \mathbb{N}$ such that

$$\mathcal{N}((T - zI)^\alpha) = \mathcal{N}((T - zI)^{\alpha+1}).$$

The dimension of $\mathcal{N}((T - zI)^\alpha)$ is the *algebraic multiplicity* of z , and the elements of $\mathcal{N}((T - zI)^\alpha)$ are the *generalized eigenvectors* of T . The dimension of $\mathcal{N}(T - zI)$ is the *geometric multiplicity* of z .

Theorem 3.9. *Let $T \in \mathcal{B}(H, H)$ be a self-adjoint operator. Then, for every $z \in \Sigma_T \setminus \{0\}$, the ascent of z is one, and the algebraic and geometric multiplicities coincide.*

4 ε -eigenvalues and compact operators

We can generalize the notion of eigenvalue in a similar manner to how we generalized the notion of spectrum by introducing the pseudo-spectrum.

Definition 4.1. Let $T \in \mathcal{B}(X, X)$ and let $\varepsilon > 0$. We call $z \in \mathbb{C}$ an ε -*eigenvalue* if there exists $x \in X$ such that

$$\|x\| = 1, \quad \|(T - zI)x\| < \varepsilon.$$

We denote by $\Sigma_{\varepsilon, T}$ the set of all ε -eigenvalues of T and call such an x an ε -eigenfunction associated with T .

Proposition 4.2. *Let $T \in \mathcal{B}(X, X)$. Then*

$$\Sigma_{\varepsilon, T} \subseteq \sigma_{\varepsilon, T}.$$

Proof. For $z \in \rho_T$, we have

$$\|R_T(z)\| = \frac{1}{\inf_{\|x\|=1} \|(T - zI)x\|}.$$

If $z \in \Sigma_{\varepsilon, T}$, then there exists $x \in X$ with $\|x\| = 1$ and

$$\|(T - zI)x\| < \varepsilon.$$

Therefore

$$\inf_{\|x\|=1} \|(T - zI)x\| < \varepsilon,$$

so

$$\|R_T(z)\| > \frac{1}{\varepsilon}.$$

Thus $z \in \sigma_{\varepsilon, T}$. □

We now focus our attention on operators with isolated spectra. Compact operators are an important class of such operators.

Proposition 4.3. *Let $T \in \mathcal{K}(X, X)$, where X is infinite-dimensional. Then*

$$0 \in \sigma_T$$

and

$$\sigma_T \setminus \{0\} = \Sigma_T \setminus \{0\}.$$

Proof. Suppose $0 \notin \sigma_T$. Then T is bijective. Since T is compact, the identity

$$I = T \circ T^{-1}$$

would be compact. This contradicts the Riesz ball theorem on an infinite-dimensional Banach space. Therefore $0 \in \sigma_T$.

Now let $z \in \sigma_T \setminus \{0\}$. Since $T - zI$ is a compact perturbation of the invertible operator $-zI$, the Fredholm alternative implies that failure of invertibility occurs precisely through a non-trivial kernel. Hence

$$\mathcal{N}(T - zI) \neq \{0\},$$

so $z \in \Sigma_T \setminus \{0\}$. □

Theorem 4.4. *Let $T \in \mathcal{K}(X, X)$ and let $\{z_n\}_{n \in \mathbb{N}} \subset \mathbb{C}$ be a sequence such that*

$$z_n \in \sigma_T \setminus \{0\}, \quad z_n \rightarrow z.$$

Then $z = 0$. Hence $\sigma_T \setminus \{0\}$ is made only of isolated points.

Proof. Since $z_n \in \sigma_T \setminus \{0\}$, the previous proposition gives $z_n \in \Sigma_T$. Choose eigenvectors x_n associated with z_n and denote by E_n the space spanned by the first n eigenvectors.

We first observe that the eigenvectors are linearly independent. Indeed, suppose by contradiction that

$$x_{n+1} = \sum_{k=1}^n \alpha_k x_k.$$

Then

$$Tx_{n+1} = \sum_{k=1}^n z_k \alpha_k x_k,$$

but also

$$Tx_{n+1} = z_{n+1}x_{n+1} = \sum_{k=1}^n z_{n+1} \alpha_k x_k.$$

Therefore

$$\sum_{k=1}^n (z_k - z_{n+1}) \alpha_k x_k = 0.$$

Since the x_k are linearly independent and $z_k \neq z_{n+1}$, all α_k vanish. This contradicts $x_{n+1} \neq 0$.

By Riesz's lemma, we can construct a sequence $\{x_n\}_{n \in \mathbb{N}}$ such that

$$\|x_n\| = 1, \quad x_n \in E_n, \quad \text{dist}(x_n, E_{n-1}) \geq \frac{1}{2}, \quad n \geq 2.$$

For $n > m \geq 2$, since

$$(T - z_m I)E_m \subseteq E_{m-1} \quad \text{and} \quad E_{m-1} \subseteq E_m \subseteq E_{n-1} \subseteq E_n,$$

we get

$$\begin{aligned} \|z_n^{-1}Tx_n - z_m^{-1}Tx_m\| &= \|z_n^{-1}(Tx_n - z_nx_n) - z_m^{-1}(Tx_m - z_mx_m) + x_n - x_m\| \\ &\geq \text{dist}(x_n, E_{n-1}) \\ &\geq \frac{1}{2}. \end{aligned}$$

If $z_n \rightarrow z \neq 0$, compactness of T implies that $\{Tx_n\}$ has a convergent subsequence. The previous inequality prevents such a subsequence from existing. Hence the only possible accumulation point is $z = 0$. $\square$

Corollary 4.5. *Let $T \in \mathcal{K}(X, X)$. Then exactly one of the following possibilities occurs:*

1. $\sigma_T = \{0\}$;
2. $\sigma_T \setminus \{0\}$ is finite;
3. $\sigma_T \setminus \{0\}$ is a sequence converging to 0.

Proof. For every integer $n \geq 1$, consider

$$\sigma_T \cap \left\{ z \in \mathbb{C} : |z| \geq \frac{1}{n} \right\}.$$

This set is either empty or finite. Otherwise there would be a subsequence converging to some non-zero point, contradicting the previous theorem. Hence the only possible accumulation point of the non-zero spectrum is 0. $\square$

Proposition 4.6. *Let $T \in \mathcal{B}(X, X)$ be a finite-rank operator, i.e.*

$$\dim \mathcal{R}(T) < \infty.$$

Then $\sigma_T \setminus \{0\}$ has finitely many elements.

Proof. The elements of $\sigma_T \setminus \{0\}$ are eigenvalues. Since $\dim \mathcal{R}(T) < \infty$, there can only be finitely many non-zero eigenvalues. $\square$

Definition 4.7. Let $\{e_n\}_{n \in \mathbb{N}} \subset H$ be a sequence of orthonormal elements of H . We say that $\{e_n\}$ is a *Hilbert basis* if

$$\overline{\text{span}\{e_n : n \in \mathbb{N}\}} = H.$$

Theorem 4.8. *Every separable Hilbert space has a Hilbert basis.*

5 Compact self-adjoint and normal operators

Theorem 5.1. *Let H be a separable Hilbert space and let $T \in \mathcal{K}(H, H)$ be self-adjoint. Then there exists a Hilbert basis composed of eigenvectors of T .*

Proof. Let $\{z_n\}_{n \geq 1}$ be the sequence of all distinct non-zero eigenvalues of T . Set

$$z_0 = 0, \quad E_0 = \mathcal{N}(T),$$

and denote by E_n the eigenspace associated with z_n . Notice that

$$\dim(E_n) < \infty$$

by the Fredholm alternative.

We prove that the E_n are mutually orthogonal. Pick $h \in E_m$ and $h' \in E_n$ with $m \neq n$. Since T is self-adjoint,

$$z_m(h, h') = (Th, h') = (h, Th') = z_n(h, h').$$

Since $z_m \neq z_n$, we obtain

$$(h, h') = 0.$$

Let

$$E := \text{span} \left(\bigcup_{n \geq 0} E_n \right).$$

We show that E is dense in H . First observe that

$$TE \subseteq E.$$

Furthermore, if $h \in E^\perp$ and $h' \in E$, then

$$(Th, h') = (h, Th') = 0,$$

so

$$Th \in E^\perp.$$

Thus

$$T(E^\perp) \subseteq E^\perp.$$

Denote by $T|_{E^\perp}$ the restriction of T to $E^\perp$. If $T|_{E^\perp}$ were not zero, then, by the previous spectral results for compact self-adjoint operators, it would have a non-zero eigenvalue. This would produce an eigenvector of T lying in $E^\perp$, contradicting the definition of E .

Therefore T vanishes on $E^\perp$, so

$$E^\perp \subseteq \mathcal{N}(T) = E_0 \subseteq E.$$

But $E^\perp$ is orthogonal to E , hence

$$E^\perp = \{0\}.$$

Thus E is dense in H .

Finally, choose a Hilbert basis in each finite-dimensional eigenspace E_n . The union of these bases is a Hilbert basis of H composed of eigenvectors of T . $\square$

Theorem 5.2. *The same result holds for compact normal operators: if $T \in \mathcal{K}(H, H)$ is normal, then there exists a Hilbert basis of H composed of eigenvectors of T .*

Theorem 5.3. *Let $T \in \mathcal{K}(H, H)$ be normal. Then*

$$\sigma_{\varepsilon, T} = \{z \in \mathbb{C} : \text{dist}(z, \sigma_T) < \varepsilon\} = \sigma_T + B_\varepsilon(0).$$

Proof. Let $\{h_n\}_{n \in \mathbb{N}}$ be a Hilbert basis formed by eigenvectors of T , with corresponding eigenvalues λ_n . If $h = \sum_n h_n(h, h_n)$, then

$$R_T(z)h = \sum_n \frac{(h, h_n)}{\lambda_n - z} h_n.$$

Consequently,

$$\|R_T(z)\| = \sup_n \frac{1}{|\lambda_n - z|} = \frac{1}{\text{dist}(z, \sigma_T)}.$$

Therefore

$$\|R_T(z)\| > \frac{1}{\varepsilon} \iff \text{dist}(z, \sigma_T) < \varepsilon,$$

which proves the claim. $\square$

Example 5.4. The components of the pseudo-spectrum describe for which values of z the Helmholtz problem is not ill-conditioned from the physical point of view. For self-adjoint or normal operators, the previous theorem gives the sharp upper bound

$$\|R_T(z)\| \leq \frac{1}{\text{dist}(z, \sigma_T)}.$$

Thus, in this setting, the degree of ill-conditioning is controlled exactly by the distance from z to the spectrum.

6 Spectral projections

Theorem 6.1. *Let $T \in \mathcal{K}(H, H)$ be normal. Fix $\mu \in \sigma_T \setminus \{0\}$, and let $\Gamma_\mu \subset \rho_T$ be a closed curve enclosing μ and no other element of σ_T . With the convention*

$$R_T(z) = (T - zI)^{-1},$$

the operator

$$\Lambda_{T, \Gamma_\mu} : H \rightarrow H, \quad \Lambda_{T, \Gamma_\mu} h := -\frac{1}{2\pi i} \int_{\Gamma_\mu} R_T(z) h \, dz$$

is the projection onto the eigenspace associated with μ .

Proof. Let $\{h_n\}_{n \in \mathbb{N}}$ be a Hilbert basis composed of eigenvectors of T , and write

$$Th_n = \lambda_n h_n.$$

The action of $R_T(z)$ can be written as

$$R_T(z)h = \sum_{n=1}^{\infty} \frac{(h, h_n)}{\lambda_n - z} h_n.$$

Thus

$$\Lambda_{T, \Gamma_\mu} h = -\frac{1}{2\pi i} \int_{\Gamma_\mu} \sum_{n=1}^{\infty} \frac{(h, h_n)}{\lambda_n - z} h_n \, dz.$$

By the Cauchy residue theorem, all terms vanish except those with

$$\lambda_n = \mu.$$

Hence

$$\Lambda_{T, \Gamma_\mu} h = \sum_{n \in I_\mu} (h, h_n) h_n, \quad I_\mu := \{n : h_n \in \mathcal{N}(T - \mu I)\}.$$

This is exactly the orthogonal projection onto the eigenspace

$$\mathcal{N}(T - \mu I).$$

□

Remark 6.2. The handwritten notes indicate that the corresponding spectral projection should also be shown in the Banach-space case.